

Full Test — Answers

Final answers with a one-line reason. For full methods, revisit the concept notes.

Objective

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- Q1** (b) **exponent**. n is the power; 2 is the base.
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- Q2** (c) **1/8**. $2^{-3} = 1/2^3$.
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- Q3** (a) **3^6** . Same base, add exponents.
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- Q4** (b) **$9/4$** . Negative power flips: $(3/2)^2$.
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- Q5** 1. Any non-zero base to the power 0 is 1.
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- Q6** **False**. $(a^2)^3 = a^6$, not a^5 .
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- Q7** (a) **$2^2 \times 3^3 \times 5$** . $540 = 4 \times 27 \times 5$.
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- Q8** (a) **Both true, R explains A**. $a^m \div a^m = a^0 = 1$.
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Short answer

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- Q9** (i) **$2^3 = 8$** (ii) **6^4** .
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- Q10** (i) **16/81** (ii) **16**.
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- Q11** (i) **7.8×10^4** (ii) **4.5×10^{-5}** .
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- Q12** **$405 = 3^4 \times 5$; $540 = 2^2 \times 3^3 \times 5$** .
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Long answer

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- Q13** (i) **$3^3 = 27$** (ii) **$p^{-7} = 1/p^7$** . ($2^{-5+6} = 3$; $3^{-10} = -7$.)
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- Q14** (i) **2^{-8}** (ii) **$(3/4)^3$** .
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- Q15** **$2^{10} = 1024$ cells**. Doubling 10 times from 1 cell.
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- Q16** **Star A is nearer**. $9.35 \times 10^6 < 6.27 \times 10^7$, since $10^6 < 10^7$.
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Total: 28 marks.